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Eliminating Retrieval Knowledge Conflicts: Cross-Validation Re-ranking with Large Language Models

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***Abstract*—In retrieval-augmented generation (RAG) systems, Large Language Models (LLMs) have been shown to be effective for re-ranking. However, existing research often prioritizes passage relevance over reliability, which can result in the incorporation of conflicting information and the generation of ambiguous responses. This issue becomes particularly pronounced when addressing inter-context knowledge conflicts, where candidate documents present contradictory information that may mislead the model. To mitigate this problem, we propose a novel cross-validation re-ranking technique designed specifically to resolve inter-context knowledge conflicts during the retrieval process. We also develop a new dataset, ContraPRT, to evaluate the ability of models to rank passages containing conflicting knowledge. Experimental results using GPT-4 and LlaMA3-70B demonstrate that our approach not only effectively filters out conflicting information but also ensures accurate passage rankings, thereby providing reliable supplementary knowledge for the generation module.**

***Index Terms*—Re-ranking, LLMs, Knowledge Conflicts, Retrieval-Augmented Generation**

I. Introduction

In recent years, with the continuous advancement of artificial intelligence technologies and the expansion of their application domains, Large Language Models (LLMs) play an increasingly important role in natural language processing. Retrieval-Augmented Generation (RAG) is a technique in natural language processing that integrates retrieval mechanisms with generative LLMs [1], [2]. By retrieving additional contextual information, it improves the accuracy and relevance of the generated text [3]–[5]

However, when integrating external information sources, issues of inter-context conflict may arise due to factors such as the authority and timeliness of the information. The advancement of RAG has exacerbated this challenge [6]–[8] . During the retrieval process, false information (e.g., fake news) and irrelevant content generated by artificial intelligence can pose significant threats [9]. Such conflicting information can result in confusion in the generated responses and lead to semantic inconsistencies in the generated content [10], [11]. Fig. 1 illustrates a specific example in which inter-context conflicts caused the LLMs to produce two unreliable answers.

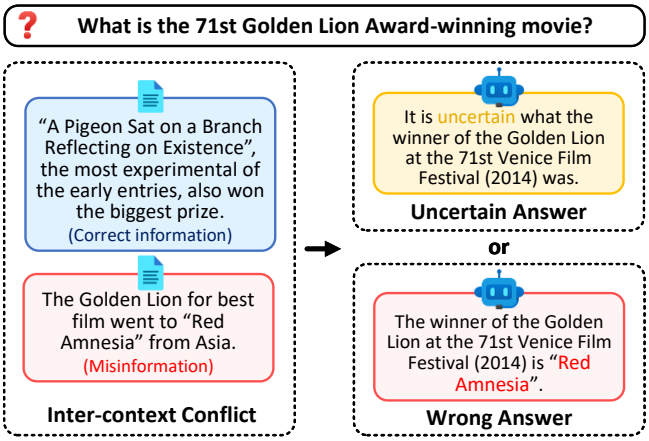


Fig. 1. An example showing how an inter-context conflict between two descriptions leads to uncertainty or incorrect answers.

Re-ranking, as the final step of the retrieval module, enhances search relevance by analyzing query intent and contextual relationships between passages [12], [13]. Additionally, the re-ranking phase integrates various signals, such as information authority and freshness, to comprehensively evaluate candidate passages. This helps filter out low-quality supplementary information that may not be fully considered during the initial retrieval.

To this end, we explore the following research questions:

- **RQ1**: *In the presence of inter-context conflicts, how can we evaluate the robustness of re-ranking methods?*
- **RQ2**: *Can LLMs, based on their inherent capabilities, filter out the correct passages from conflicting information?*
- **RQ3**: *How can we develop a more effective re-ranking*

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method to resolve inter-context conflicts during the retrieval process?

To address **RQ1**, we construct a query-ranking dataset containing conflicting knowledge to evaluate the performance and reliability of the re-ranking module. Specifically, we process documents from the ContraDoc dataset [14], design corresponding query questions, and build a new dataset, ContraPRT (Passage Ranking Task), tailored for evaluating passage-level ranking in conflict scenarios.

To explore **RQ2**, we investigate whether LLMs can rely on their inherent reasoning abilities to effectively handle conflicts. Our findings suggest that current LLMs struggle to appropriately filter and rank conflicting information. When faced with complex supplementary knowledge, contradictions within the context can lead to logical inconsistencies, resulting in unstable ranking outcomes.

To address **RQ3**, we propose a cross-validation re-ranking method to resolve inter-context knowledge conflicts. Our approach leverages the semantic consistency and relevance of the candidate passage set to filter out conflicting information [15]. As illustrated in Fig. 2, our method integrates both pairwise and listwise ranking strategies through a "detect-select-rank" process. This enables the system to identify and eliminate conflicts detected during retrieval. Experimental results demonstrate that our method effectively selects the correct passages from conflicting pairs while removing disruptive data, thereby enhancing the overall robustness of the RAG system.

II. Related Work

A. Inter-context Conflicts in LLMs

Knowledge conflicts refer to the encounter of contradictory or inconsistent information during the retrieval process. In RAG systems, the challenges posed by knowledge conflicts are particularly significant, as these systems rely on passages retrieved from large databases. However, due to uncertainties such as data sources and data quality, misinformation, including fake news [16], [17] or AI-generated false information [18], may be introduced, leading to various issues [19].

Existing language models are highly vulnerable to misinformation attacks often generating content that includes contradictory or incorrect information [20], [21]. Furthermore, conflicting information can introduce biases in the generated content, making the output lean towards inaccurate or biased perspectives.

Reference [11] demonstrate that LLMs struggle to distinguish between true and false information. Handling specific conflicts, such as opinion-based conflicts, presents even more severe challenges [14]. When a model identifies conflicting information, it may adjust the confidence level of the generated answers, presenting different responses depending on the information [6]. However, the value of these generated answers significantly decreases, as users cannot obtain definitive answers from the output [22], [23].

The key to resolving conflict issues lies in obtaining more reliable information segments from conflicting data through

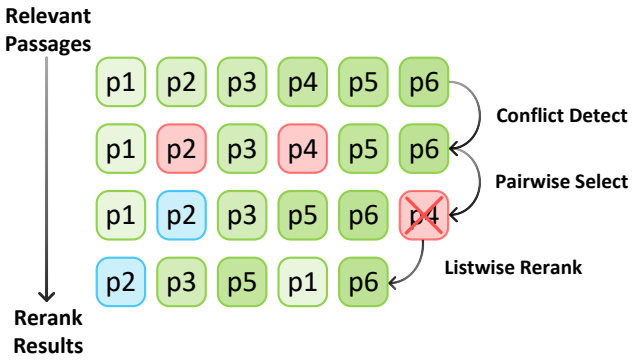


Fig. 2. The proposed multi-stage re-ranking framework eliminates inter-context knowledge conflicts through conflict detection, pairwise filtering, and listwise re-ranking.

a series of evaluations and validations. Researchers have proposed various methods to mitigate the impact of inter-context conflicts. For example, one approach involves detecting misinformation in text by integrating external tools [24] or using LLMs to generate weak labels based on predefined credibility signals from the input text, and then aggregating these labels through weak supervision techniques to predict the authenticity of the input [14]. Reference [9] also employ query enhancement techniques to retrieve additional fragments that assess the credibility of the answers.

Current strategies for resolving inter-context conflicts primarily rely on the internal knowledge of models or external tools. However, in specialized domains, accurate information is often not available through Web searches and model knowledge alone is insufficient to resolve all conflict issues. Therefore, designing an efficient method to address inter-context conflicts remains a significant challenge.

B. Re-ranking with LLMs

Re-ranking techniques are divided into supervised and unsupervised methods. Supervised methods, which rely on large amounts of annotated data for training, face challenges such as high annotation costs, scalability issues, and difficulties in environments where annotated data is scarce [25]–[27]. These methods also struggle with generalization when exposed to new data that differs from the training set, often leading to overfitting. As a result, performance may be strong on familiar data but poor on new, complex, or diverse ranking tasks [28]. In recent years, researchers have begun using LLMs for text re-ranking [29]–[31] and their effectiveness has been validated in multiple experiments [32], [33]. Recent studies have also explored issues related to fairness and positional bias when using LLMs for ranking [34], [35]. This highlights that LLMs, when employed as re-ranking agents, can consider a broader range of factors.

LLM-based re-ranking methods primarily include listwise and pairwise strategies. The listwise method seeks to maximize performance metrics by optimizing the order of the entire list [29], [36]. This approach considers not only relevance but also diversity and other factors. However, due to the need

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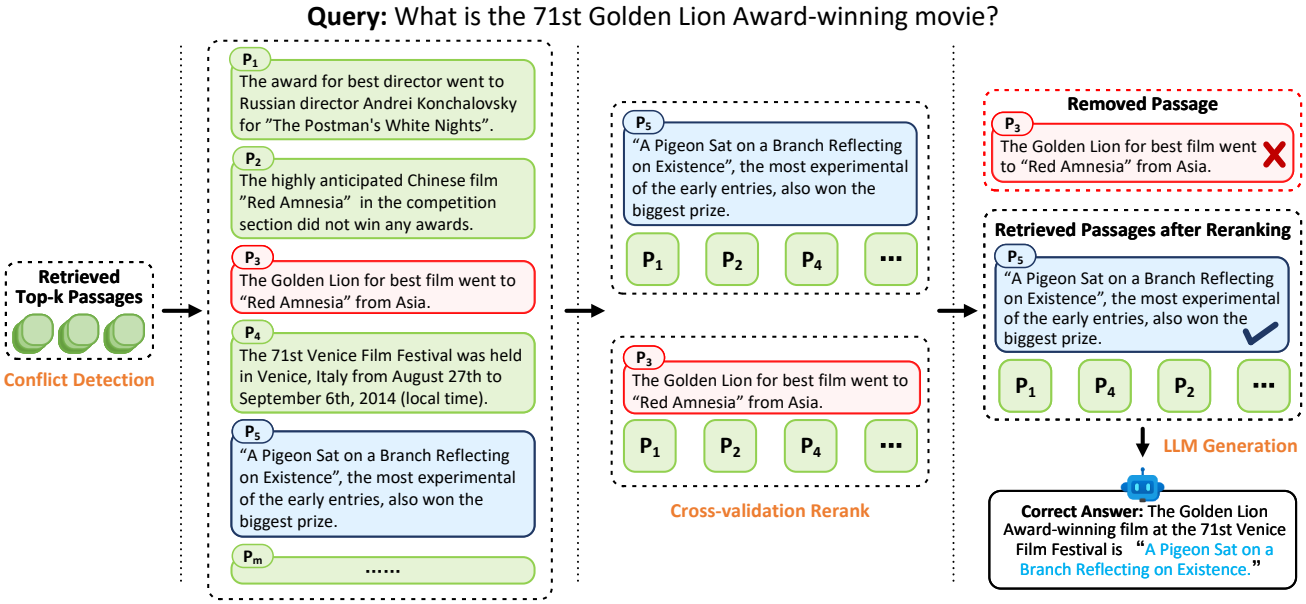


Fig. 3. An example of re-ranking using cross-validation to eliminate inter-context conflicts. The information retrieval process begins with the query 'What is the 71st Golden Lion Award-winning movie?' In the conflict detection stage, the system identifies conflicting information, followed by cross-validation re-ranking to select reliable passage and discard inaccuracies, resulting in the final ranking of accurate information.

to process the entire set of candidate passages, it involves high computational complexity, which may not effectively resolve conflicts. In practical applications, especially in complex scenarios, the listwise method based on LLMs sometimes produces disorganized ranking results.

On the other hand, the pairwise method simplifies the ranking process by transforming it into a series of pairwise comparisons, making it more intuitive and manageable [30]. When the set of candidate passages is small, this method can provide more precise ranking results, aiding in the accurate identification of the most relevant passages for answering [37]. However, the pairwise method primarily focuses on local ordering and may fail to capture global ranking features. This limitation becomes particularly evident in scenarios involving knowledge conflicts, where the pairwise method struggles to detect conflicting information within the candidate passage set.

III. Method

Fig. 3 illustrates the detailed process of our proposed method. We utilize the processing capabilities of LLMs to detect and filter inter-context conflict information. To more effectively address these conflicts, we introduce a cross-validation method to compare and select conflicting information during the re-ranking process. Ultimately, this approach produces a passage set that excludes conflicting knowledge, ensuring that the LLMs generate accurate and reliable responses, thus minimizing the risk of uncertain or incorrect answers.

Formally, given a user query Q and a set of information passages $P = \{p_1, p_2, \ldots, p_m\}$, we assume that p_1 and p_2 contain conflicting information. We then utilize the LLMs M to execute the following steps:

1) **Cross-validation:** The model analyzes the content of p_1 and p_2 to identify their conflicts and compares these passages against the rest of the set $\{p_3, p_4, \ldots, p_m\}$ to evaluate their individual relevance and accuracy.
2) **Passage Selection:** Based on this evaluation, the model identifies which of the passages, p_1 or p_2, is less accurate or relevant and designates it as p_{false}, while retaining the more accurate or relevant passage as p_{correct}.
3) **Re-ranking:** The model removes p_{false} from the candidate set and re-ranks the remaining passages $\{p_{\text{correct}}, p_3, \ldots, p_m\}$, producing an updated ranking result using LLMs M.

A. Inter-context Conflict Detection

After the initial retrieval, we obtain a set of candidate passages related to the query, each labeled with a unique identifier (e.g., [1]). These passages are sequentially input into the LLMs, which are tasked with identifying passages that contain conflicting information relevant to the intent of query. If such passages are detected, the model outputs the identifiers of the conflicting passage pairs, such as [2] and [5]. This detection process requires the model to account for global information across the candidate passage set and the contextual relationships between passages, making this step listwise in nature. It is important to emphasize that the final selection focuses on the passage pair with the most evident inter-context conflict.

B. Eliminating Inter-context Conflict

In the filtering phase, where conflicting passage pairs are addressed, we evaluate the semantic knowledge of the remaining passages against that of the conflicting pairs. By

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TABLE I
STATISTICS OF CONTRAPRT.

Attribute	Count
Number of Queries	200
Conflict Type Statistics	
Content	126
Emotion/Mood/Feeling	28
Negation	43
Perspective/View/Opinion	33
Causal	5
Relation	9
Factual	26
Numeric	47
Document Type Statistics	
Story	41
News	65
Wiki	94
Chunk Length (Avg.)	251.42 (chars)
Document Length (Avg.)	4976.19 (chars)

determining which passage demonstrates greater consistency and relevance with the rest of the passage set, we can identify the more reliable one. This approach, termed cross-validation, involves a comprehensive comparison of semantic coherence and logical alignment. The process of semantic extraction and reasoning required for cross-validation necessitates advanced language processing capabilities, which are well-supported by the rapid advancements in LLMs. Consequently, we apply the cross-validation method to powerful LLMs, directing them to select passages that are contextually aligned with the query and to discard those containing contradictory information. This method produces a refined set of candidate passages, ensuring that the resulting set is free from conflicting information that could compromise the quality of the generated answers. This step is grounded in pairwise comparisons, specifically aimed at resolving conflicting passage pairs.

Following this filtering process, we instruct the LLMs to re-rank the remaining passages based on their relevance, prioritizing those most closely aligned with the query. This re-ranking step employs a listwise approach, ensuring that the final ranked set is optimized for accuracy and contextual relevance.

IV. EXPERIMENTS

A. Dataset and Metrics

Existing passage ranking benchmarks overlook inter-context knowledge conflicts, despite their frequent occurrence in real-world retrieval results. To address this, we develop ContraPRT based on ContraDoc [14], a new benchmark designed to evaluate models' abilities to resolve such conflicts. The dataset includes 200 documents, divided into passage chunks, with corresponding query questions. The design follows the format of TREC 2019 tasks [38], a widely recognized benchmark in retrieval tasks.

a) Data Construction: We first extract key conflicting information from the conflict sentences marked in the source dataset through a human-machine collaboration framework. Based on this conflicting information, we generate corresponding query questions. The dataset includes 200 queries, each of which has been manually verified to ensure its high relevance to the conflicting information in the documents. During the document segmentation process, we specifically ensure that conflicting sentences are placed in different chunks to prevent internal knowledge confusion within the same chunk. Additionally, taking into account the window size of LLMs and the semantic integrity of passages, we divide the documents into chunks of approximately 250 characters each. This segmentation method not only maintains the semantic coherence of the text but also ensures that the model can effectively reason within its processing capabilities. Through this meticulous construction process, the ContraPRT dataset provides a reliable evaluation platform for passage ranking in the presence of knowledge conflicts. Table I presents the statistics of ContraPRT, including the number of queries, conflict type statistics, document type statistics, and length statistics. It is important to note that a set of conflicting passages may involve multiple conflict types.

b) Evaluation Metrics: For relevance evaluation, we adopt the *Normalized Discounted Cumulative Gain* (nDCG@k), which considers both relevance scores and ranking positions. The Discounted Cumulative Gain (DCG@k) is calculated as:

$$\text{DCG@k} = \sum_{i=1}^{k} \frac{rel_i}{\log_2(i+1)} \tag{1}$$

where rel_i is the relevance score of the i-th chunk in the ranking. The normalized version is defined as:

$$\text{nDCG@k} = \frac{\text{DCG@k}}{\text{IDCG@k}} \tag{2}$$

where IDCG@k is the DCG value under an ideal ranking. A higher nDCG@k indicates better relevance ranking quality.

$$\text{Error@k} = \frac{\sum_{j=1}^{n} \mathbb{I}\left(\max_{i\in[1,k]} \mathbb{I}(\text{error}_{i,j})\right)}{n} \tag{3}$$

where n denotes the total number of ranking tasks, and $\mathbb{I}(\text{error}_{i,j})$ is an indicator function that equals 1 if the i-th chunk in the j-th task contains factual errors, and 0 otherwise. The inner maximum function detects whether any of the top-k results contain an erroneous chunk, and the final metric reports the average error rate across all tasks.

To assess the impact of re-ranking on generation quality in retrieval-augmented settings, we evaluate the factual consistency of responses generated using the top-5 re-ranked passages as external knowledge. The assessment measures the proportions of correct, hallucinated, and uncertain answers. Uncertain answers refer to cases where conflicting retrieved information prevents the model from providing a definitive response (e.g., "Due to conflicting information, I cannot provide a clear answer"), which is not counted as a hallucination.

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TABLE II
RESULTS (NDCG AND TOP-K ERROR) ON CONTRAPRT. BEST PERFORMING ARE MARKED BOLD.

Method	nDCG@1	nDCG@5	nDCG@10	Error@1	Error@5	Error@10
RankGPT4	0.515	0.760	0.751	0.170	0.750	0.850
RankGPT3.5-turbo	0.600	0.753	0.732	0.410	0.885	0.960
Cohere Rerank-english-v3.0	0.492	0.710	0.703	0.285	0.915	0.950
Llama3-70B	0.665	0.792	0.778	0.190	0.780	0.890
GPT-3.5-turbo w/ cross-validation **(ours)**	0.670	0.787	0.769	0.075	0.250	0.305
GPT-4 w/ cross-validation **(ours)**	0.761	0.831	**0.819**	0.030	0.190	0.190
Llama3-70B w/ cross-validation **(ours)**	**0.770**	**0.833**	0.817	**0.005**	**0.055**	**0.075**

TABLE III
EVALUATION OF FACTUAL CONSISTENCY. BEST PERFORMING RESULTS ARE MARKED IN BOLD.

Method	Uncertain (%)	Correct (%)	Hallucination (%)
RankGPT4	51.5	14.5	34.0
RankGPT3.5-turbo	52.0	26.0	22.0
Cohere Rerank-English-v3.0	55.0	16.5	28.5
GPT-3.5-turbo w/ cross-validation **(ours)**	22.0	69.0	9.0
GPT-4 w/ cross-validation **(ours)**	14.0	82.0	4.0
Llama3-70B w/ cross-validation **(ours)**	**10.0**	**88.5**	**1.5**

B. Baselines

We select the following baselines for comparison:

- **RankGPT:** We adopt the re-ranking method described by [29] and applie it to three models: GPT3.5-turbo, GPT4 [39], and Llama3-70B [40]. Importantly, due to the enhanced context window support in LLMs, we did not use the sliding window strategy.
- **Cohere Rerank3[1]:** We utilize cohere rerank-english-v3.0 model. Cohere Rerank3 is a newly developed foundation model specifically designed for efficient enterprise search and retrieval.

We evaluate our proposed method on three LLMs: GPT-3.5-Turbo, GPT-4, and Llama3-70B. The GPT series models have demonstrated strong performance across a wide range of tasks; however, researchers must consider the cost associated with API calls when utilizing these models. In contrast, Llama3-70B is an open-source model, enabling researchers to use it without significant concerns about cost. This accessibility facilitates more extensive experimentation and development, particularly in academic or non-commercial settings.

C. Main Results

The main results are presented in the Table II. Overall, our method effectively eliminates inter-context knowledge conflicts during the ranking process. We observe the following results:

- As shown by the nDCG scores in Table II, the Cohere model struggles in reranking, particularly in the Top-1 scenario, highlighting the limitations of supervised methods. In the Top-1/5/10 scenarios, our proposed method achieves the best results on GPT-4 and Llama3-70B, with nDCG@{1,5,10} scores of 0.770, 0.833, and 0.819, respectively, marking significant improvements over previous re-ranking approaches.
- Additionally, we analyze the number of disruptive passages in the ranking results. We observe that conventional re-ranking methods failed to prevent the inclusion of conflicting knowledge. In the Top-10 scenario, more than 85% of incorrect passages were selected. However, after applying cross-validation re-ranking, the proportion of incorrect conflicting passages significantly decreased. This improvement is most notable with Llama3-70B, where the error rate in the Top-10 scenario drops to 7.5%, and in the Top-1 scenario, only one incorrect passage is selected.
- Even with the less capable GPT-3.5-Turbo model, our proposed method substantially reduces the error rate, with nDCG scores approaching those of the more powerful language models

These results demonstrate the effectiveness of our cross-validation re-ranking approach in managing knowledge conflicts across different scenarios and model capacities, contributing to the development of more robust RAG systems capable of handling complex informational contexts. After cross-validation, LLMs are also able to provide reasoned explanations based on logical inferences.

Furthermore, we evaluate the impact of our re-ranking method on factual consistency, as shown in Table III. Results show that the cross-validation approach significantly improves response reliability in RAG systems. Llama3-70B achieves 88.5% correctness with only 10.0% uncertain and 1.5% hallucinated answers, while GPT-4 reaches 82.0% correctness with a 4.0% hallucination rate. In contrast, traditional baselines like RankGPT4 and Cohere Rerank yield over 50% uncertain answers and hallucination rates of 34.0% and 28.5%, respectively. Even GPT-3.5-Turbo benefits, with correctness

[1]https://cohere.com/blog/rerank-3.

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TABLE IV
RESULTS (NDCG AND TOP-K ERROR) BASED ON DIFFERENT METHOD.

Method	nDCG@1	nDCG@5	nDCG@10	Error@1	Error@5	Error@10
Llama3-70B w/ prompt indicating conflict	0.487	0.675	0.634	0.110	0.655	0.835
Llama3-70B w/ cross-validations **(ours)**	0.770	0.833	0.817	**0.005**	**0.055**	**0.075**
GPT4 w/ prompt indicating conflict	0.525	0.757	0.749	0.400	0.810	0.905
GPT4 w/ cross-validations **(ours)**	0.761	0.831	**0.819**	0.030	0.190	0.190

rising from 26.0% to 69.0% and hallucinations dropping from 22.0% to 9.0%. These results demonstrate the effectiveness of our method in reducing uncertainty and hallucination across different models.

D. Ablation studies

a) The Effectiveness of Cross-validation Method: To evaluate the effectiveness of the cross-validation method in filtering conflicting information, we compare the experimental results of LLMs using two different prompt designs. Specifically, we add additional instructions to the re-ranking prompt template, informing the large language model that the candidate passage set contains conflicting information. We then instruct the model to use its knowledge to identify and filter out this conflicting information, requiring it to produce a ranking result that excludes passages containing such conflicts.

The results of the ablation experiments are shown in Table IV. From these results, we observe that even with the additional prompt, the large language model struggles to effectively manage passages with conflicting information. As a result, the re-ranking process becomes less coherent, leading to a decrease in the reliability of the outcomes.

After adding prompts to inform the model of inter-context conflicts between passages, the results for nDCG@{1,5,10} are affected, with most scores showing a decline. In particular, in the Top-5 and Top-10 sequences, many passages still contain conflicting information. This suggests that relying solely on the knowledge and logical reasoning abilities of models to filter out conflicting information is insufficient to improve the reliability of the ranking results.

b) Conflict Types and Quantities: To further explore the impact of conflicting information on the re-ranking performance of LLMs, we differentiate between scenarios involving a single conflict type and those involving multiple conflict types. For example, some conflicting information pertains only to "numerical" conflicts, while others involve multiple types simultaneously, such as "negation," "relationships," and "content." We assess the ability of models to handle these scenarios based on the number of conflict types present in the data. As shown in Table V, although the number of conflict types increases, the number of selected passages containing conflicting information does not change significantly.

In addition, we categorize the conflict types and evaluate the effect of different conflict aspects on performance. Table VI shows that when confronted with simple conflicts, such as "negation," the model can effectively filter out the correct passages. However, in more complex conflict scenarios, such as those involving opinions or emotional conflicts, the error rate increases.

TABLE V
ERROR RATES FOR DIFFERENT MODELS WITH CROSS-VALIDATION, COMPARING CASES WITH ONE AND MULTIPLE CONFLICT TYPES.

Model	Types	Error@10	Error@5	Error@1
GPT-3.5-turbo	= 1	0.297	0.228	0.079
GPT-3.5-turbo	≥ 2	0.313	0.273	0.071
GPT-4	= 1	0.178	0.178	0.020
GPT-4	≥ 2	0.202	0.202	0.040
Llama3-70B	= 1	0.109	0.109	0.010
Llama3-70B	≥ 2	0.091	0.051	0.010

c) Additional Relevant Information: During the cross-validation process, the model may lack sufficient supplementary knowledge due to low contextual relevance, resulting in re-ranking outputs that still contain inter-contextual knowledge conflicts.

To determine the amount of supplementary contextual information required for the large language model to make accurate decisions during re-ranking, we conducted additional experiments on the erroneous ranking tasks. We introduced 1 to 3 relevant information segments into the candidate set to aid the model in improving its decision-making. These supplementary passages were closely related to the query and semantically aligned with the correct passages.

Fig. 4 presents the results. In certain simpler conflict cases, supplementing the model with one or two highly relevant information passages enables it to reason more effectively using the additional knowledge, allowing for a more accurate determination of which passage aligns more closely with objective facts, thereby yielding more reliable re-ranking outcomes.

Even after supplementing with three relevant information segments, some ranking errors persist, mainly due to conflicting perspectives. We hypothesize that individuals may hold differing views on a topic in real world. For these tasks, the model tends to include all passages in the re-ranking results, treating the conflicting information as an objective representation of diverse viewpoints.

V. DISCUSSION

a) Impact of Knowledge Conflicts: We test the responses of three models on the query when the Top-5 returned

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TABLE VI
ERROR RATES FOR DIFFERENT CONFLICT ASPECTS USING GPT-4 WITH CROSS-VALIDATION. PVO DENOTES PERSPECTIVE/VIEW/OPINION, AND EMF REPRESENTS EMOTION/MOOD/FEELING.

Aspect	Error@10	Error@5	Error@1
Negation	0.047	0.047	0.000
Content	0.187	0.187	0.033
Causal	0.200	0.200	0.000
PVO	0.242	0.242	0.061
Numeric	0.255	0.255	0.021
EMF	0.321	0.321	0.107
Relation	0.375	0.375	0.000

TABLE VII
THE PROPORTION OF HALLUCINATIONS GENERATED BY THE MODEL WHEN THE TOP-5 PASSAGES CONTAIN CONFLICTING INFORMATION.

Model	Hallucination Rate (%)
GPT-3.5-turbo	64.00
Llama3-70B	38.00
GPT-4	26.00

passages contain knowledge conflicts. The results are shown in Table VII. We observe that GPT-3.5-turbo generates hallucinations, producing factually incorrect content in 64% of queries. In comparison, the hallucination rates for Llama3-70B and GPT-4 are 38% and 26%, respectively. These findings demonstrate that knowledge conflicts significantly affect the accuracy of model outputs, with models being more prone to generating erroneous or ambiguous responses, particularly in complex conflict scenarios or less common knowledge domains.

b) Robustness of RAG: As an increasing amount of data is applied in the field of RAG, building a more robust RAG system becomes crucial. If the external knowledge contains conflicts, the model may generate answers that lack reference value, thereby undermining the benefits of retrieval enhancement. To address this issue, it is essential to improve the filtering capability of the retrieval module so that it can identify more reliable information sources. Additionally, verifying content consistency through multi-perspective validation and implementing post-processing steps before generation are critical. This approach increases the accuracy of the generated answers meanwhile enhances trust in the outputs of system.

b) Effectiveness of Re-ranking: Re-ranking serves as a critical bridge between the retrieval and generation modules, playing a vital role in ensuring the quality and accuracy of the generated content. In RAG systems, the re-ranking process effectively filters the most appropriate knowledge inputs by assessing the relevance and credibility of the retrieved information. This not only improves the reliability of the information but also significantly reduces misunderstandings and errors arising from knowledge conflicts. Thus, re-ranking not only improves

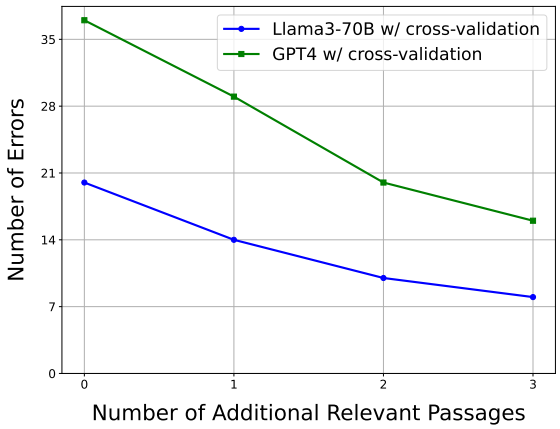


Fig. 4. The effect of additional information on cross-validation for GPT-4 and Llama3-70B.

the precision and effectiveness of information selection but also significantly enhances the system's ability to handle complex queries and meet diverse informational needs, further strengthening the robustness and adaptability of RAG system.

VI. CONCLUSION

In this paper, we address the challenges and limitations of current re-ranking methods in RAG systems, particularly in handling information conflicts. We introduced a dataset, ContraPRT, to evaluate these methods against inter-context conflicts. Our findings underscore the need for improvements, leading us to propose a new cross-validation re-ranking method using LLMs, which significantly enhances conflict resolution.

Our method demonstrates strong performance not only on advanced models like GPT-4 but also on the open-source Llama3-70B, providing researchers with cost-effective alternatives to commercial API. The proposed method enhances the robustness and reliability of RAG systems in complex informational environments and lays the groundwork for future advancements in intelligent RAG system design. This research contributes to resolving knowledge conflicts in RAG systems and paves the way for further studies.

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REFERENCES

[1] K. Guu, K. Lee, Z. Tung, et al., "Retrieval augmented language model pre-training," in Proceedings of the International Conference on Machine Learning, 2020, pp. 3929–3938.

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[2] P. Lewis, E. Perez, A. Piktus, et al., "Retrieval-Augmented generation for Knowledge-Intensive NLP tasks," in Advances in Neural Information Processing Systems, 2020, vol. 33, pp. 9459–9474.

[3] K. Shuster, S. Poff, M. Chen, et al., "Retrieval Augmentation Reduces Hallucination in Conversation," in Findings of the 2021 Conference on Empirical Methods in Natural Language Processing, 2021, pp. 3784–3803.

[4] J. Xu, A. Szlam, J. Weston, "Beyond Goldfish Memory: Long-Term Open-Domain Conversation," in Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics, Volume 1: Long Papers, 2022, pp. 5180–5197.

[5] S. Siriwardhana, R. Weerasekera, E. Wen, et al., "Improving the Domain Adaptation of Retrieval Augmented Generation (RAG) Models for Open Domain Question Answering," Transactions of the Association for Computational Linguistics, vol. 11, 2023, pp. 1–17.

[6] H.-T. Chen, M. Zhang, and E. Choi, "Rich knowledge sources bring complex knowledge conflicts: Recalibrating models to reflect conflicting evidence," in Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, 2022, pp. 2292–2307.

[7] J. Xie, K. Zhang, J. Chen, R. Lou, and Y. Su, "Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts," in Proceedings of the Twelfth International Conference on Learning Representations, 2023.

[8] K. Wu, E. Wu, and J. Zou, "How faithful are RAG models? Quantifying the tug-of-war between RAG and LLMs' internal prior," arXiv preprint arXiv:2404.10198, 2024.

[9] K. Pelrine, A. Imouza, C. Thibault, M. Reksoprodjo, C. Gupta, J. Christoph, J.-F. Godbout, and R. Rabbany, "Towards reliable misinformation mitigation: Generalization, uncertainty, and GPT-4," in Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, 2023, pp. 6399–6429.

[10] S. Wu, J. Xie, J. Chen, T. Zhu, K. Zhang, and Y. Xiao, "How easily do irrelevant inputs skew the responses of large language models?," arXiv preprint arXiv:2404.03302, 2024.

[11] Z. Jin, P. Cao, Y. Chen, K. Liu, X. Jiang, J. Xu, Q. Li, and J. Zhao, "Tug-of-war between knowledge: Exploring and resolving knowledge conflicts in retrieval-augmented language models," arXiv preprint arXiv:2402.14409, 2024.

[12] Y. Xi, J. Lin, W. Liu, X. Dai, W. Zhang, R. Zhang, R. Tang, and Y. Yu, "A bird's-eye view of reranking: From list level to page level," in Proceedings of the Sixteenth ACM International Conference on Web Search and Data Mining, 2023, pp. 1075–1083.

[13] M. Glass, G. Rossiello, M. F. M. Chowdhury, A. Naik, P. Cai, and A. Gliozzo, "Re2G: Retrieve, Rerank, Generate," in Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, 2022, pp. 2701–2715.

[14] J. Li, V. Raheja, D. Kumar, "Contradoc: Understanding Self-Contradictions in Documents with Large Language Models," in Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1: Long Papers, 2024, pp. 6509–6523.

[15] L. Hagström, D. Saynova, T. Norlund, M. Johansson, and R. Johansson, "The effect of scaling, retrieval augmentation and form on the factual consistency of language models," in Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, 2023, pp. 5457–5476.

[16] Y. R. Fung, K.-H. Huang, P. Nakov, and H. Ji, "The battlefront of combating misinformation and coping with media bias," in Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2022, pp. 4790–4791.

[17] B. Hu, Q. Sheng, J. Cao, Y. Shi, Y. Li, D. Wang, and P. Qi, "Bad actor, good advisor: Exploring the role of large language models in fake news detection," in Proceedings of the AAAI Conference on Artificial Intelligence, 2024, pp. 22105–22113.

[18] C. Chen and K. Shu, "Can LLM-generated misinformation be detected?," in Proceedings of the Twelfth International Conference on Learning Representations, 2023.

[19] J. A. Leite, O. Razuvayevskaya, K. Bontcheva, and C. Scarton, "Detecting misinformation with LLM-predicted credibility signals and weak supervision," arXiv preprint arXiv:2309.07601, 2023.

[20] E. Kortukov, A. Rubinstein, E. Nguyen, and S. J. Oh, "Studying large language model behaviors under realistic knowledge conflicts," arXiv preprint arXiv:2404.16032, 2024.

[21] L. Pan, W. Chen, M.-Y. Kan, and W. Y. Wang, "Attacking open-domain question answering by injecting misinformation," in Proceedings of the 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (Volume 1: Long Papers), 2023, pp. 525–539.

[22] L. Gao, Z. Dai, P. Pasupat, A. Chen, A. T. Chaganty, Y. Fan, V. Zhao, N. Lao, H. Lee, D.-C. Juan, et al., "RARR: Researching and revising what language models say, using language models," in Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2023, pp. 16477–16508.

[23] N. F. Liu, T. Zhang, and P. Liang, "Evaluating verifiability in generative search engines," in Findings of the Association for Computational Linguistics: EMNLP 2023, 2023, pp. 7001–7025.

[24] I. Chern, S. Chern, S. Chen, W. Yuan, K. Feng, C. Zhou, J. He, G. Neubig, P. Liu, et al., "FacTool: Factuality detection in generative AI–A tool augmented framework for multi-task and multi-domain scenarios," arXiv preprint arXiv:2307.13528, 2023.

[25] X. Ma, L. Wang, N. Yang, F. Wei, and J. Lin, "Fine-tuning LLaMA for multi-stage text retrieval," arXiv preprint arXiv:2310.08319, 2023.

[26] R. Nogueira, W. Yang, K. Cho, and J. Lin, "Multi-stage document ranking with BERT," arXiv preprint arXiv:1910.14424, 2019.

[27] J.-H. Ju, J.-H. Yang, and C.-J. Wang, "Text-to-text multi-view learning for passage re-ranking," in Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, 2021, pp. 1803–1807.

[28] Z. Peng, X. Wu, Q. Wang, S. Rajanala, and Y. Fang, "Q-PEFT: Query-dependent parameter efficient fine-tuning for text reranking with large language models," arXiv preprint arXiv:2404.04522, 2024.

[29] W. Sun, L. Yan, X. Ma, S. Wang, P. Ren, Z. Chen, D. Yin, and Z. Ren, "Is ChatGPT good at search? Investigating large language models as re-ranking agents," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 14918–14937, 2023.

[30] Z. Qin, R. Jagerman, K. Hui, et al., "Large language models are effective text rankers with pairwise ranking prompting," in Findings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics, 2024, pp. 1504–1518.

[31] X. Ma, X. Zhang, R. Pradeep, and J. Lin, "Zero-shot listwise document reranking with a large language model," arXiv preprint arXiv:2305.02156, 2023.

[32] R. Pradeep, S. Sharifymoghaddam, and J. Lin, "Rankvicuna: Zero-shot listwise document reranking with open-source large language models," arXiv preprint arXiv:2309.15088, 2023.

[33] Y. Ma, Y. Cao, Y. Hong, and A. Sun, "Large language model is not a good few-shot information extractor, but a good reranker for hard samples!," in Findings of the Association for Computational Linguistics: EMNLP 2023, 2023, pp. 10572–10601.

[34] Y. Wang, X. Wu, H.-T. Wu, Z. Tao, and Y. Fang, "Do large language models rank fairly? An empirical study on the fairness of LLMs as rankers," arXiv preprint arXiv:2404.03192, 2024.

[35] R. Tang, C. Zhang, X. Ma, J. Lin, and F. Türe, "Found in the middle: Permutation self-consistency improves listwise ranking in large language models," in Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), 2024, pp. 2327–2340.

[36] S. Yoon, E. Lee, J. Kim, Y. Kim, H. Yun, and S.-w. Hwang, "ListT5: Listwise reranking with fusion-in-decoder improves zero-shot retrieval," arXiv preprint arXiv:2402.15838, 2024.

[37] N. B. Shah and M. J. Wainwright, "Simple, robust and optimal ranking from pairwise comparisons," J. Mach. Learn. Res., vol. 18, no. 199, pp. 1–38, 2018.

[38] N. Craswell, B. Mitra, E. Yilmaz, D. Campos, and E. M. Voorhees, "Overview of the TREC 2019 deep learning track," arXiv preprint arXiv:2003.07820, 2020.

[39] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat, et al., "GPT-4 technical report," arXiv preprint arXiv:2303.08774, 2023.

[40] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, et al., "LLaMA: Open and efficient foundation language models," arXiv preprint arXiv:2302.13971, 2023.

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